dp练习2

人员

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上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/3302

王向东老师周日三点半C++dp练习

刷新

#	用户名	姓名	编程分	时间	A (11 / 22)	B (11 / 25)	C (9 / 12)	D (7 / 10)	E (3 / 16)
1	liuqi	刘祺	5	13:2:38	(-3) 0:38:21	1:18:51	1:52:57	4:56:27 ♥	(-2) 2:36:2
2	liuliwei	柳力玮	5	111:28:33	(-1) 0:24:31 ♥	0:36:33 ♥	1:25:26	53:49:17	(-3) 53:52:46
3	tianxinyi	田心一	4	6:8:30	0:30:40	1:0:12	1:24:34 ♥		(-4) 1:53:4 ♥
4	gaojianhuan	高健桓	4	172:42:34	(-2) 0:36:32	1:24:16	1:52:44	168:9:2	
5	jibohan	纪博涵	4	174:59:10	(-2) 0:47:48	1:35:43	(-1) 2:4:17	(-3) 168:31:22	
6	liuchenxi	刘宸熙	4	175:51:10	(-3) 0:48:36	(-5) 1:43:33	2:3:11	168:35:50	
7	yangyongcheng	杨咏丞	4	255:58:4	0:38:26	(-2) 1:36:5	(-2) 126:10:14	126:13:19	
8	chujinyang	初锦阳	3	4:1:30	0:30:53	1:33:35	1:57:2		(-2)
9	yuanzhao	苑钊	3	5:59:58	0:45:10	(-5) 1:45:14	1:49:34		(-2)
10	liruihan	李瑞涵	3	170:32:4	0:30:41	1:46:26		168:14:57	
11	jiangshuzhang	蒋叔璋	2	2:33:29	0:31:36	(-2) 1:21:53			

本周作业

https://cppoj.kids123code.com/contest/3411 (课上讲了 A ~ C 题, 课后作业是 C 题必做)

课堂表现

同学们今天课上做题表现整体不是太好, 课下一定要把今天的题目都补完。

课堂内容

租用游艇 (上周作业)

f[i]: 到位置 i 所需要的最少租金

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 200 + 5;
int w[maxn][maxn], f[maxn];

int main()
{
```

```
  int n; cin >> n;
  for (int i = 1; i <= n; ++i) {
    for (int j = i+1; j <= n; ++j) cin >> w[i][j];
  }

  memset(f, 0x3f, sizeof(f));
  f[1] = 0;
  for (int i = 2; i <= n; ++i) {
    for (int j = 1; j <= i-1; ++j) f[i] = min(f[i], f[j] + w[j][i]);
  }
  cout << f[n] << endl;
  return 0;
}
```

跑步

f[i][j]: 一共跑了 i 圈, 最后一次跑了 j 圈时, 一共有多少合法方案

```
#include <bits/stdc++.h>

using namespace std;

typedef long long LL;
const int maxn = 500 + 5;
LL f[maxn][maxn];

int main()
{
  int n; cin >> n;
  f[0][0] = 1;
  for (int i = 1; i <= n; ++i) {
    for (int j = 1; j <= i; ++j) {
      for (int k = 0; k < j; ++k) {
        f[i][j] += f[i-j][k];
      }
    }
  }

  LL res = 0;
  for (int i = 0; i <= n; ++i) res += f[n][i];
  cout << res - 1 << endl;
  return 0;
}
```

最长括号匹配

f[i]: 以 i 结尾时, 最长的合法括号子串长度是多少

```cpp
#include <bits/stdc++.h>

using namespace std;

const int maxn = 1e6 + 5;
char s[maxn];
int f[maxn];

int main()
{
  cin >> (s+1);
  int n = strlen(s+1);

  int pos = 0;
  for (int i = 1; i <= n; ++i) {
    if (s[i]=='(' || s[i]=='[') continue;

    int len = f[i-1];
    int p = i - len - 1;
    if (s[i] == ')') {
      if (s[p] == '(') f[i] = len+2+f[p-1];
    } else {
      if (s[p] == '[') f[i] = len+2+f[p-1];
    }

    if (f[i] > f[pos]) pos = i;
  }

  int l = pos - f[pos] + 1, r = pos;
  for (int i = l; i <= r; ++i) cout << s[i];
  cout << endl;
  return 0;
}
```

签到题

先求出来 以每个数结尾 和 以每个数开头 时最长连续上升串的长度

然后枚举修改哪个数即可

```cpp
#include <bits/stdc++.h>

using namespace std;

const int maxn = 1e6 + 5;
int w[maxn];
int f[maxn];  // f[i]: 以 i 结尾的最长连续上升子串的长度
int s[maxn];  // s[i]: 以 i 开头的最长连续上升子串的长度

int main()
```

```
{
  int n; cin >> n;
  for (int i = 1; i <= n; ++i) cin >> w[i];

  f[1] = 1;
  for (int i = 2; i <= n; ++i) {
    if (w[i] > w[i-1]) f[i] = f[i-1] + 1;
    else f[i] = 1;
  }

  s[n] = 1;
  for (int i = n-1; i >= 1; --i) {
    if (w[i] < w[i+1]) s[i] = s[i+1] + 1;
    else s[i] = 1;
  }

  int res = 0;
  for (int i = 1; i <= n; ++i) res = max(res, f[i]);

  for (int i = 2; i <= n; ++i) res = max(res, f[i-1]+1);

  for (int i = 1; i <= n-1; ++i) res = max(res, s[i+1]+1);

  for (int i = 2; i <= n-1; ++i) {
    if (w[i+1] - w[i-1] >= 2) res = max(res, f[i-1]+1+s[i+1]);
  }
  cout << res << endl;
  return 0;
}
```